



ZIMBABWE SCHOOL EXAMINATIONS COUNCIL
General Certificate of Education Advanced Level

WOOD TECHNOLOGY AND DESIGN

Paper 3 Drawing and Design

6027/3

3 hours

SPECIMEN PAPER

Additional materials:

- Computer with Autocad applications
- Printer /Plotter
- USB flash / DVD / CD
- A4 Drawing paper
- Calculator

INSTRUCTIONS TO CANDIDATES

Type your centre number, candidate number and name in the spaces provided on all your electronic answer sheets.

INFORMATION FOR CANDIDATES

The total marks for this paper is 100.

Marks are given in [] brackets at the end of each question or part question.

At the end of the examination save your work on the **USB flash** and **print your work**.

Fasten your work securely together.

This specimen paper consists of 4 printed pages.

Copyright: Zimbabwe School Examinations Council, Specimen paper.

SECTION A (20 MARKS)

*Answer **all** questions.*

*You are advised to spend not more than **30 minutes** on this section.*

- 1 (a) State any **two** disadvantages of CAD. [2]
- (b) Name any **two** types of 3D printers. [2]
- (c) Distinguish between a line and a polyline. [2]
- (d) Give any **two** disadvantages of Computer Aided Manufacturing (CAM). [2]
- (e) What is the use of Object Snap. [2]
- (f) The diagram in **Figure 1** shows a stool in 3D. [10]

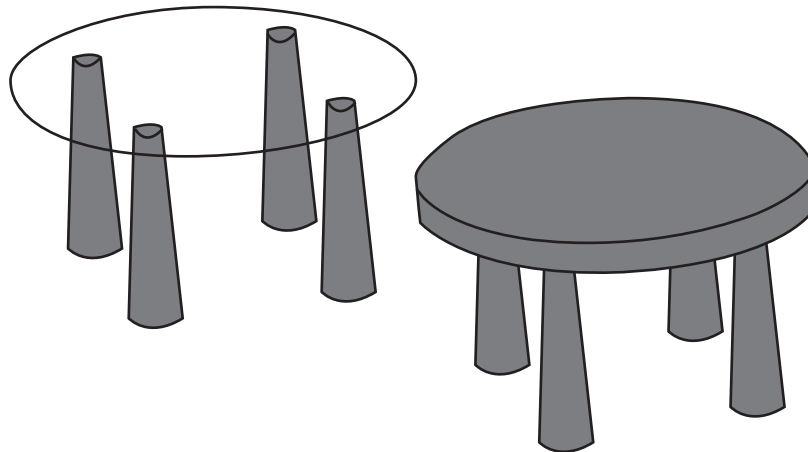


Fig. 1

Using suitable dimensions of your choice reproduce the stool.

SECTION B: (80 MARKS)

Answer **one** question from this section. You are advised to spend **2 hours 30 minutes** on this section.

- 2 **Figure 2** shows a multi-purpose Hall with a retractable flat roof.

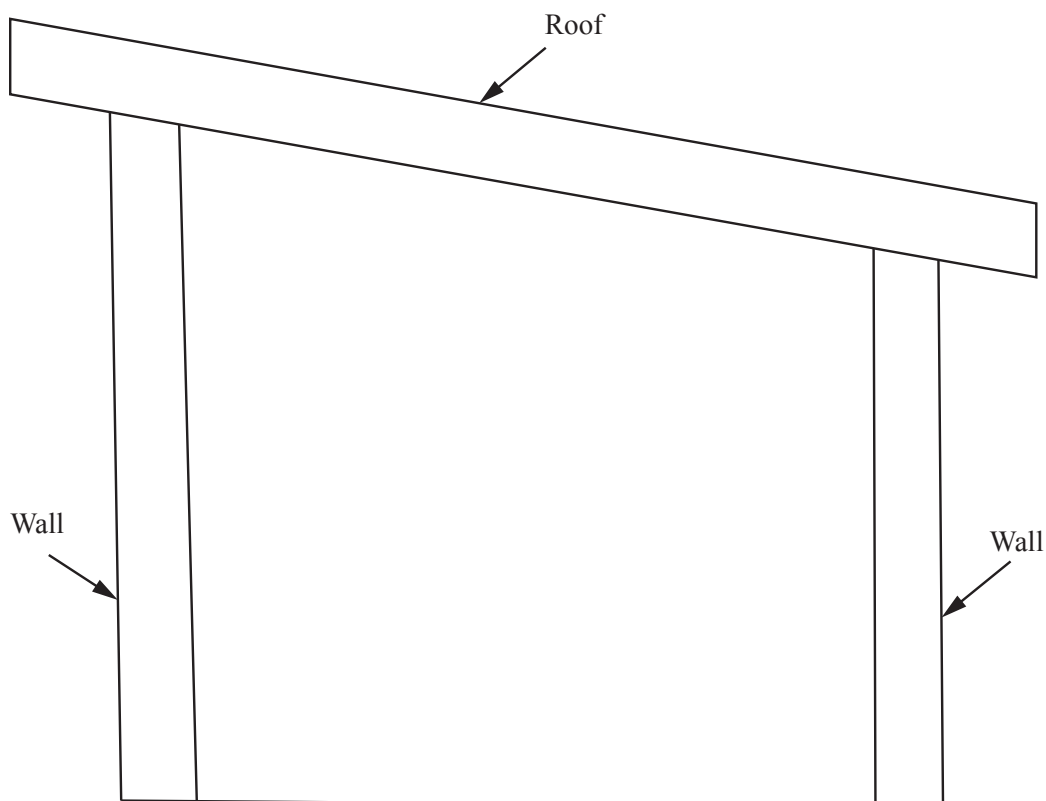


Fig. 2

- (a) Design in 3D, **two** methods of opening and closing the roof when necessary. [50]
- (b) Show the type of truss or roof support structure to be used on the roof design. [15]
- (c) Show a method of ensuring that the retractable roof is storm proof. [10]
- (d) (i) Name any **two** materials that can be used in the construction of the retractable roof. [2]
- (ii) State any **three** mechanical properties of the materials named in (i) above. [3]

- 3** At a local college, there is a bathroom for both able-bodied learners and wheelchair bound ones. A cabinet has to be fitted and has to be mounted in such a way that it can be raised or lowered to the required height.
- (a)
 - (i) Design in **3D**, a mounted bathroom cabinet that can be used by the learners in the college mentioned above. [25]
 - (ii) Show in **3D**, any **two** mechanisms/methods that can be used to raise or lower the cabinet. [25]
 - (iii) Show in **3D**, the method of securing or locking the mounted cabinet in a particular position. [16]
 - (b)
 - (i) State any **three** possible materials that can be used to make the cabinet. [3]
 - (ii) Explain how the cabinet will be finished in relation to the different possible materials you have stated in **(b)(i)**. [3]
 - (iii) Apply a chamfer on the outside edges of the cabinet you have designed. [2]
 - (c) Show a method of fixing a mirror on the bathroom cabinet. [6]